Inferring Hidden Enzymatic Activity from Mutated DNA

Lion Bremeyer
2025400578
Tsinghua Shenzhen International Graduate School
Shenzhen, China
lion-silas@bremeyer.de

Daniel Lorand Szabo
2025400575
Tsinghua Shenzhen International Graduate School
Shenzhen, China
daniel.szabo@tum.de

***Abstract*—Activation-Induced Cytidine Deaminase (AID) is the key enzyme in B-cell receptor maturation during adaptive immune response. Although its biological role is well established, its kinetic mechanisms remain mostly elusive. In this project, we propose a latent probabilistic modeling framework that treats AID activity as a hidden stochastic process along the B-cell receptor gene sequence. We model the enzyme using a Markovian model, with two enzyme activity states (bound/ unbound) for each base as hidden states, assuming observable mutations happen dependent on enzyme state, base state, and motif bias of enzyme activity. Using likelihood maximization, we estimate biologically interpretable parameters, calculate the posterior probabilities for enzyme states and use the Viterbi algorithm to reconstruct the most likely hidden enzyme activity path from observed mutation data. This project helps to better understand the probabilistic nature of AID dynamics during immune responses.**

I. Introduction

A. Biological Motivation

1) Adaptive Immunity and Germinal Centers: The adaptive immune system distinguishes itself from the innate immune system by its ability to respond flexibly to almost every possible antigen it may encounter. This ability depends heavily on the vast diversity of potential antibodies that B-cells are able to produce. One major contributor to this diversity is the V(D)J recombination during B-cell development in the bone marrow, during which B-cell receptor (BCR) genes segments are recombined in random configurations to form complete and functional BCR genes. This process generates a diverse initial primary BCR repertoire. However, it is insufficient for high-affinity pathogen neutralization and does not allow for real-time adaptation. To achieve the necessary specificity, upon meeting an unknown antigen, active B-cells migrate to specialized structures located within the lymph nodes called germinal centers. Here they undergo affinity maturation, an evolutionary process of mutation and selection. Central to this process is Somatic Hypermutation (SHM), which introduces point mutations into variable regions of the BCR genes at a rate of ca. 10^{-3} base pairs (bp) per generation. This is 10^6 times the spontaneous rate in somatic cells.

2) Enzymatic Mechanism of AID: The reason for this elevated mutation rate is the Activation-Induced Cytidine Deaminase (AID). Biologically, AID acts as a DNA-editing enzyme that acts specifically on single-stranded DNA (ssDNA) which is exposed during transcription [1]. The enzyme preferably attacks the variable (*IGV*) and switch (*IgS*) regions of the immunoglobulin (BCR) loci [2]. AID catalyzes the hydrolytic deamination of the exocyclic amine at position 4 of the cytidine ring, thereby converting desoxycytidine (dC) to desoxyuridine (dU). As uracil is normally only present in RNA, this conversion creates a U:G mismatch in DNA [2]. Importantly, AID does not bind the DNA uniformly, but actually exhibits a high preference for specific local sequences called motifs. The enzyme targets cytidines with higher probability that are contained in so called ”hotspots”, most notably the WRC sequence (where W = Adenine/Thymine, R = Purine [A/G], and C = Cytidine) [3]. Similarly, AID shows a reduced probability to catalyze deamination when encountering a ”coldspot” motif such as the SYC sequence (S = Guanine/Cytosince, Y = Thymine/Cytosine, and C = Cytidine) [4]. This motif preference is likely due to structural fit of the ssDNA into the catalytic center of the enzyme [3].

3) Downstream Processing and Mutation: The mutations that are observable in sequencing data are, however, not the direct product of AID but rather the outcome of the cell's attempt to repair the AID-induces U:G mismatches. The repair of an uracile in DNA relies on the recruitment of error-prone DNA repair pathways. There are three major outcomes after AID-catalyzed deamination [5]:

- **Base Excision Repair (BER):** The enzyme Uracil-DNA Glycosylase can excise the uracil base which leaves a so called abasic site. The Translesion Synthesis Polymerase can then bypass this site inserting the missing base. This however, is highly prone to error, frequently leading to both transitions and transversions (C→G or C→A).
- **Mismatch Repair (MMR):** Alternatively, the MSH2/MSH6 complex can recognize the U:G mismatch and excise a large patch of DNA in the vicinity of the site. The subsequent gap filling by the error-prone polymerase η, frequently introduces mutations at A:T base pairs nearby, which spreads the mutation radius beyond the original site where the deamination occurred.
- **Replication:** It is also possible that the presence of uracil is not detected in time to be repaired before the replication machinery arrives. The high-fidelity DNA-

polymerase will insert an adenine opposite to the uracil (essentially treating it like a thymine). This leads to a permanent C→T transition in the daughter strand.

4) Kinetics of AID: While the chemical reaction of AID is quite well understood already, the spatial dynamics of AID are still somewhat elusive. This is also due to its highly dynamic nature. The enzyme does not simply bind its substrate, catalyze once and dissociate, but instead there is evidence for it to be processive [1]. Once bound to DNA, it can slide, hop or diffuse along the strand to catalyze multiple deaminations in a single binding event [6]. This processivity leads to clusters of mutations and implies that the probability of a mutation at any given site is dependent on the enzyme's spatial position [7]. This kinetic behavior is the central justification for using a Hidden Markov Model (HMM) to model its activity along the DNA. The physical states Unbound (searching) and Bound (scanning with stochastic point events of deamination) are not directly observable in static sequencing data. They are hidden variables with a stochastic and context dependent nature along the sequence. By modeling these dynamics explicitly, we aim to infer parameters such as the enzyme's dwell time and binding probability also accounting for motif bias.

II. Related Work

Due to its central role in B-cell maturation, the Activation-Induced Cytidine Deaminase has been and still is being studied extensively. Some publications focus on elaborating its structural and biochemical properties. Other work is dedicated to the computationally modeling its mutational footprint. This section focuses on some of the existing literature on AID's scanning mechanism, the role of the sequence context and existing mathematical approaches to model Somatic Hypermutation.

1) Scanning and Processivity: Early studies have already noted, that AID does not act on single stranded DNA in a "hit-and-run" manner, but rather functions processively. Once bound to ssDNA, it can catalyze multiple deaminations within a short window, before dissociating again. Biochemical analysis suggest that this processivity is governed by a stochastic scanning mechanism, where the enzyme moves along the DNA via a combination of sliding and jumping. Mak et al. (2013) [8] tried to model this scanning dynamic mathematically. For this, they formulated a random walk model to analyze AID's movement with the help of a master equation approach in order to interpret mutational data from reporter cassettes (synthetically introduced observable DNA fragments). Their work suggests that AID act on ssDNA bidirectionally with a "dwell time" of approximately 2.7 seconds per motif. They also show that it has an average contact length of roughly 18 nucleotides. Recent work also showed that AID acts specifically on the ssDNA exposed by the transcription bubbles, with AID interacting directly with the transcription elongation machinery, such as RNA Polymerase II and the factor Spt5 [9].

2) Sequence Context: While the preference of AID for WRC hotspots is well documented, very recent work also indicates that its activity is influenced by a more complex sequence grammar that regards more than the immediate 3-nucleotide motif. Bartl et al. (2025) [10] demonstrate that identical WRC motifs exhibit varying mutation frequencies depending on where they are located in the gene. They suggest that this variation is due to a motif-specific sequence context that influences how AID is interacting with the ssDNA and potentially affecting the enzyme's binding stability. This sequence dependence is also supported by findings suggesting that AID contains a specificity loop that differs from other enzymes of the same family, potentially being more affine to purine-rich regions [3].

3) Computational Models: To quantify the context biases, several compuational models have been proposed. The most widely used is the S5F (Synonymous, 5-mer, Functional) model which was published by Yaari et al. (2013) [11]. By analyzing synonymous mutations in high-throughput sequencing data, the S5F model estimates how likely a mutation and substitution is in every possible 5-mer motif. This proved to sucessfully account for the motif bias (such as WRC hotspot and SYC coldspot) and serves as a baseline for simulating SHM patterns. The limitations of k-mer based models, such as S5F, lies in their inability to capture long range dependencies or the specific position of a motif within the gene. To address this, Tang et al. (2021) [12] developed "DeepSHM". This is a deep learning model which uses Convolutional Neural Networks (CNNs). With it, they showed that extending the sequence context from 5-mers to 15-mers or 21-mers can improve the prediction of mutation frequencies. While these models are effective at predicting mutation rates, they generally treat mutations as independent events conditioned on sequence context. They fail, however, to account for the latent activity of AID as it scans, binds and catalyzes along the DNA. This shows a gap for this project to address by directly modeling AID activity as a hidden path.

III. Methodology

The starting point for this project are 100 synthetically generated DNA sequences each of length 1000 nucleotides. We sample from a discrete uniform distribution with nucleotides A, T, C, G as possible outcomes. As AID works on single-stranded DNA, we only synthesize one of the two strands of DNA. In section III-A we show how the AID-mutagenesis process is modeled, i. e. how the hidden enzyme state sequences and final mutated DNA sequences corresponding to the original DNA sequences are created and what parameters we then need to recover later. It is important to note that the underlying reason for using synthetic data is to validate the HMM's capabilities as the ground truth hidden states for evaluation are only known for synthetically generated data. In section III-B we discuss the implemented likelihood maximization method to recover the model parameters. We use the ground truth model parameters to retrieve information about the hidden states: In section III-C we discuss how the posterior probabilities for the hidden states are calculated using the forward-backward algorithm and in section III-D

we present the application of Viterbi's algorithm to recover the most likely binding trajectory (i.e. hidden states).

A. Generative HMM to model AID-Mutagenesis

Our implemented HMM has two hidden states: unbound/background (state 0) and bound/scanning (state 1). Each state transition from position t to $t+1$, with $t = 0, \ldots, T-1$ and sequence length $T = 1000$, occurs with different probabilities: AID recruitment probability ($0 \to 1$) is described by α_{recruit} and scanning persistence / processivity ($1 \to 1$) is implemented as λ_{scan}. Emissions (i.e. the actual mutations) are conditioned on the hidden state.

The hidden states are defined by a binary space $S = \{0, 1\}$, where S_t is the binding status of the enzyme at the nucleotide in position t. $S_t = 0$ corresponds to the enzyme being in the unbound state representing the regions where it is not attached to the DNA. $S_t = 1$ is the bound or scanning state, where the enzyme binds the ssDNA as substrate and is capable of catalysis. The transition between these states is governed by a stationary transition matrix A, where $A_{ij} = P(S_t = j | S_{t-1} = i)$. These transition rates are derived from two biological kinetic rates: **Recruitment** ($\alpha_{recruit}$) represents the probability that the enzyme initiates ssDNA binding from the background state ($0 \to 1$). **Processivity** (λ_{scan}) is the probability that the enzyme, once bound, remains attached to the ssDNA and translocates to the subsequent base ($1 \to 1$). Therefore, the transition matrix is given by:

$$A = \begin{pmatrix} 1 - \alpha_{\text{recruit}} & \alpha_{\text{recruit}} \\ 1 - \lambda_{\text{scan}} & \lambda_{\text{scan}} \end{pmatrix}$$

With these parameters, a high λ_{scan} results in longer continuous stretches of state 1 which simulates the sliding of AID on ssDNA, while $\alpha_{recruit}$ controls the frequency of binding events.

We defined the emission probabilities $P(O_t | S_t)$ as observing a nucleotide O_t that differs from the germline (unmutated) base G_t, given the hidden enzyme state S_t at position t. If the enzyme is unbound ($S_t = 0$) it is also inactive. Thus, all observed mutations in $S_t = 0$ arise solely from sequencing error which we model as a uniform noise process. The observed base O_t differs from G_t only with fixed probability ϵ (noise), distributed uniformly amongst all three non-germline bases.

$$P(O_t \mid S_t = 0) = \begin{cases} 1 - \epsilon & \text{if } O_t = G_t \\ \frac{\epsilon}{3} & \text{if } O_t \neq G_t \end{cases} \tag{1}$$

If the enzyme, however, is bound ($S_t = 1$), AID activity is modeled by a two-step stochastic process. A catalytic firing event followed by a mutation realization. The probability of a firing event $p_{fire}(t)$ is non-zero only at cytosines ($G_t = C$) and is modulated by two upstream bases ($t-2, t-1$). We define the firing probability as:

$$p_{\text{fire}}(t) = p_{\text{base}} \times \begin{cases} b_{\text{hot}} & \text{if } G_{t-2..t} \in \text{WRC} \\ b_{\text{cold}} & \text{if } G_{t-2..t} \in \text{SYC} \\ 1 & \text{otherwise} \end{cases} \tag{2}$$

Parameter	Value	Role
α_{recruit}	0.03	AID recruitment (0→1)
λ_{scan}	0.95	scanning persistence (1→1)
p_{base}	0.04	baseline C activation
b_{hot}	10.0	WRC hotspot bias
b_{cold}	0.1	SYC coldspot bias
η	0.8	mutation realization
ϵ	10^{-4}	background noise

TABLE I: Parameters of the two-state AID Hidden Markov Model used for sequence mutagenesis.

where p_{base} is the baseline activation rate, and b_{hot} and b_{cold} are the bias multipliers. If a firing event occurs, it results in an observable C $\to$ T mutation with efficiency η. This the effective mutation probability is $p_{\text{mut}}(t) = p_{\text{fire}}(t) \cdot \eta$. The emission probability in $S_t = 1$ is therefore:

$$P(O_t \mid S_t = 1) = \begin{cases} p_{\text{mut}}(t) & \text{if } G_t = \text{C}, O_t = \text{T} \\ 1 - p_{\text{mut}}(t) & \text{if } G_t = \text{C}, O_t = \text{C} \\ 1 & \text{if } G_t \neq \text{C}, O_t = G_t \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

B. Parameter Inference with Likelihood Maximization

In order to recover the model parameters θ we rely on the so called forward algorithm [13]. We can set up the likelihood function for each observation. In the next step, we minimize the negative-log-likelihood of the whole dataset (sum of individual likelihoods). The objective function that we set up is:

$$\hat{\theta} = \arg\max_{\theta} \sum_i \log \sum_{S^{(i)}} P\Big(O^{(i)} \mid G^{(i)}, S^{(i)}, \theta\Big)\, P\Big(S^{(i)} \mid \theta\Big),$$

where $\hat{\theta}$ are the estimated model parameters, $O^{(i)}$ are the observed mutated sequences and $G^{(i)}$ are the input sequences. In this equation we sum over all hidden state paths $S^{(i)}$ that could generate the observation sequence. The marginalization of $S^{(i)}$ is only feasible thanks to dynamic programming in the forward algorithm which is possible as a direct consequence of the Markov property. We use the SciPy implementation of the L-BFGS-B numerical optimization method to find minima [14] [15]. The initialization of L-BFGS-B can have significant impact on search direction, finding minima, and overall convergence. [16]. Here we choose the starting point for the first run $x_0 = \big(\alpha_{\text{recruit}}, \lambda_{\text{scan}}, p_{\text{base}}, b_{\text{hot}}, b_{\text{cold}}, \eta\big) = (0.05,\ 0.50,\ 0.05,\ 1.0,\ 1.0,\ 0.50)$ and for the subsequent reruns (in total five runs) sampling and box constraints as shown in table II.

C. Posterior State Probabilities with Forward–Backward Algorithm

The prerequisites from hereon are known and fixed model parameters $\hat{\theta}$. These can come from the likelihood maximization as discussed in section III-B or, as we do in this project, be taken as known (i. e. we assume we found the true model parameters). The second approach allows us to look at the estimation of the posterior state probabilities as a separate

Param.	Bounds	Restart sampling
α_{recruit}	$(10^{-6}, 1-10^{-6})$	$\text{Uniform}(10^{-6}, 1-10^{-6})$
λ_{scan}	$(10^{-6}, 1-10^{-6})$	$\text{Uniform}(10^{-6}, 1-10^{-6})$
p_{base}	$(0, 1)$	$\text{Uniform}(0, 1)$
b_{hot}	$(10^{-3}, 200)$	$\text{LogUniform}(10^{-3}, 200)$
b_{cold}	$(10^{-3}, 200)$	$\text{LogUniform}(10^{-3}, 200)$
η	$(0, 1)$	$\text{Uniform}(0, 1)$

TABLE II: Multi-start initialization scheme for robustness to local optima. Log-uniform sampling is a natural choice for positive scale parameters (bias multipliers).

and independent sub-problem. We implement the forward-backward algorithm [13], but in contrast to the cited work [13], we only use it to estimate the posterior probability (and not to estimate model parameters). The forward variables are $\alpha_t(s) = \log P(O_{0:t}, S_t = s \mid G, \hat{\theta})$ and backward variables are $\beta_t(s) = \log P(O_{t+1:T-1} \mid S_t = s, G, \hat{\theta})$. Both are computed via dynamic programming. Since the observation subsequences $O_{0:t}$ and $O_{t+1:T-1}$ together form the full observation sequence O, their product in probability space yields

$$P(O, S_t = s \mid G, \hat{\theta}).$$

By definition of conditional probability,

$$P(S_t = s \mid O, G, \hat{\theta}) = \frac{P(O, S_t = s \mid G, \hat{\theta})}{P(O \mid G, \hat{\theta})}.$$

Substituting the forward and backward expressions gives

$$P(S_t = s \mid O, G, \hat{\theta}) = \frac{\exp(\alpha_t(s) + \beta_t(s))}{P(O \mid G, \hat{\theta})}.$$

Because the denominator does not depend on s, the posterior state probability simplifies to

$$P(S_t = s \mid O, G, \hat{\theta}) \propto \exp\big(\alpha_t(s) + \beta_t(s)\big).$$

Normalization across states $s \in \{0, 1\}$ at each position yields independent posterior state estimates along the sequence.

D. Viterbi Decoding for Hidden State Recovery

The task of decoding describes the problem: Given as input the model parameters of the HMM and a sequence of observations (ground truth and mutations), find the most probable sequence of hidden states. We solve this task using the Viterbi algorithm [13]. As we have done in the previous section we restrict ourselves to looking at the decoding task as an independent sub-problem with known model parameters that correspond to the true values. As with the forward algorithm in the likelihood calculations in section III-B we use dynamic programming for the Viterbi algorithm. The foundation of the calculations are the Viterbi score variables:

$$v_t(s) = \max_{S_{0:t-1}} \log P\Big(O_{0:t}, S_t = s \mid G, \hat{\theta}\Big).$$

We initialize them as

$$v_0(0) = \log P(O_0 \mid S_0 = 0, G), \qquad v_0(1) = -\infty,$$

as we now from out assumption that we start in the hidden state 0. For each next sequence position the new Viterbi score is calculated by the recursion

$$v_t(s) = \log P(O_t \mid S_t = s, G) + \max_{s' \in \{0,1\}} \Big[v_{t-1}(s') + \log P(S_t = s \mid S_{t-1} = s')\Big].$$

Additionnaly, at each step we calculate the backpointer $b_t(s)$ for each state s, a vector that stores the most likely path of how we arrived at position t. It is computed as

$$b_t(s) = \arg\max_{s'} \left[v_{t-1}(s') + \log P(S_t = s \mid S_{t-1} = s')\right].$$

Once the end of the sequence $T-1$ is reached, we terminate the Viterbi algorithm

$$S^*_{T-1} = \arg\max_s v_{T-1}(s),$$

and return the most likely hidden state paths via backtracking $b_t(s)$. The Viterbi algorithm gives us a single representative trajectory of hidden states.

IV. Results

A. Synthetic Data Generation

To validate the HMM framework described in III-A, we first generated a synthetic dataset using the generative model with fixed biological priors. The simulation produced 100 sequences, each of length $T = 1000$ nucleotides.

The generated dataset contained a total of 1152 latent firing events, resulting in 928 observable mutations after accounting for the mutation efficiency ($\eta \approx 0.8$). This yields an average mutation frequency of roughly 9.28 mutations per sequence, which reflects the natural sparsity of somatic hypermutation. Approximately 37.8% (37,810 bound positions out of 100,000 total bases) were denoted with the hidden state $S_t = 1$, which shows the processivity of AID scanning ($\lambda_{scan} = 0.95$).

B. Model Parameter Recovery

We performed Maximum Likelihood Estimation (MLE) to infer the model parameters θ solely from the observed mutated sequences. We used the L-BFGS-B numerical optimization algorithm with 5 random restarts to avoid local minima. The best fit achieved a negative log-likelihood (NLL) of 3219.86.

Table III presents the comparison between the ground truth parameters used for generation and the recovered values. The model successfully recovered the spatial kinetic parameters: The recruitment probability ($\hat{\alpha}_{recruit} \approx 0.038$) and the scanning processivity ($\hat{\lambda}_{scan} \approx 0.938$) were close to their true values (relative errors of +26% and -1.2%). This indicates that the HMM correctly inferred the average dwell time and binding frequency of the enzyme.

However, the recovery of the emission parameters revealed significant issues. As shown in Table III, the mutation efficiency was underestimated ($\hat{\eta} \approx 0.32$), while the hotspot bias was drastically overestimated ($\hat{b}_{hot} \approx 70.6$).

Consequently, while the model is able to predict where the enzyme binds (based on the good estimates for processivity

TABLE III: Comparison of Ground Truth vs. Recovered Parameters

Parameter	Truth	Recovered	Rel. Error
$\alpha_{recruit}$	0.03	0.0378	+26.0%
λ_{scan}	0.95	0.9381	-1.2%
p_{base}	0.04	0.1072	+168.0%
b_{hot}	10.0	70.648	+606.5%
b_{cold}	0.1	0.0725	-27.5%
η	0.8	0.3238	-59.5%

and recruitment), the specific probabilities of firing versus mutation seem to be numerically unstable. To decouple this optimization error from the evaluation of the decoding algorithms, we utilize the true ground truth parameters for the hidden state inference analysis in the following section.

C. Hidden State Inference and Decoding

To evaluate the model's ability to recover the latent AID binding profiles, we assume the model parameters are known ($\hat{\theta} \approx \theta_{true}$) and focus on quantifying the accuracy of the Viterbi decoding and the Forward-Backward posterior probabilities.

1) Binding Profiles: We first investigated the model's behavior on individual sequences. Fig. 1 shows the ground truth binding path against the inferred Viterbi path and the posterior probability $P(S_t = 1|O)$ for the sequence with the highest number of mutations. The Viterbi path exhibits very few state changes which results in prolonged streaks of either bound or unbound prediction. This fails to recover the true hidden path with high resolution, but gives a conservative estimate of which regions the enzyme is active in. Comparatively, the posterior with threshold results in smaller streaks of predicted state 1, which resembles the hidden path more accurately. Aggregating across all sequences, Fig. 2 shows the per-position occupancy. The trend of the true hidden state mean (dashed black line) is visible in the mean posterior probability (blue line), demonstrating that the Forward-Backward algorithm can serve as an estimator of the global binding occupancy. The lower graph of Fig. 2 shows the aggregated mutation occurrences across all sequences with only three noise derived mutations visible.

2) Classification Performance: We treated the recovery of the hidden state at each nucleotide position as a binary classification problem. Fig. 3 presents the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.

The model achieves an ROC Area Under the Curve (AUC) of 0.718, indicating better performance than random guessing (0.5). The Precision-Recall AUC is 0.668. At a decision threshold of 0.5, the model exhibits a precision of 73.2% and a recall of 39.2% (F1-score = 0.51). This rather conservative behavior with high precision but moderate recall is a common phenomenon when dealing with sparse emission signal. The model requires strong evidence (mutations) to confidently assign the "Bound" state, often missing silent binding events where no deamination occurred.

3) Segmentation Accuracy: Beyond point-wise classification, we assessed the quality of the recovered segments using the Dice coefficient ($2 \cdot \frac{\text{Area of Overlap}}{\text{Sum of Areas}}$) and the IoU metric ($\frac{\text{Intersection}}{\text{Union}}$). Fig. 4 displays the distribution of these metrics across the 100 sequences. The histograms show a bimodal distribution. A subset of sequences achieves high overlap (Dice > 0.6), corresponding to cases where mutation clusters are well-defined. However, the Viterbi path (orange) tends to also yield sequences with low overlap. Here, the algorithm may default to the background state (State 0) because of mutational sparsity for the entire sequence. The "Posterior hard" method (thresholding $P(S_t = 1) > 0.5$) shows a slightly more robust distribution. This is likely due to its tendency to allow more and smaller stretches of singular state prediction.

V. Discussion

A. Challenges in Parameter Estimation

A central motivation for using synthetic data in our project was to establish a ground truth for evaluation. In real biological sequencing, the hidden states (enzyme binding) are never directly observable, making it impossible to quantify the accuracy of our inference.

Our analysis revealed significant challenges in recovering the model parameters via Maximum Likelihood Estimation (MLE). Unlike standard HMMs, where the Baum-Welch algorithm (Expectation-Maximization) offers closed-form updates for the parameters, our emission probabilities depend on a product of multiple factors: baseline activation, motif-specific biases (b_{hot}, b_{cold}), and mutation efficiency (η). This structure prevents simple analytic updates, which forced us to resort to direct numerical minimization of the Negative Log-Likelihood (NLL).

While the algorithm successfully recovered the kinetic parameters ($\alpha_{recruit}, \lambda_{scan}$), the emission parameters turned out to be unstable and we were not able to recover them properly. As observed in the results, the optimizer converged to a local minimum with unrealistically high hotspot bias and low mutation efficiency. This happens because the likelihood function for a sparse signal (only ≈ 9 mutations per 1000 base pairs) is relatively flat. So, the effective signal-to-noise ratio is low, making it difficult for the optimizer recover coupled, hard to estimate, parameters.

B. Viterbi Decoding vs. Posterior Probabilities

The Viterbi algorithm, which searches the single most likely path, turned out to be highly conservative. Because state transitions are penalized by the transition matrix, Viterbi basically requires a cluster of mutations to justify change to the bound state. This explains its high precision but low recall. It correctly identifies regions where the enzyme was definitely active but misses scanning events that did not lead to a mutation. It seems to be suitable to identify regions of high enzymatic activity.

In contrast, the posterior probabilities (derived from the Forward-Backward algorithm) is a rather soft measure of enzyme occupancy. By thresholding these probabilities, we get better segmentation metrics (higher IoU and Dice scores). This suggests that the posterior distribution captures the stochastic

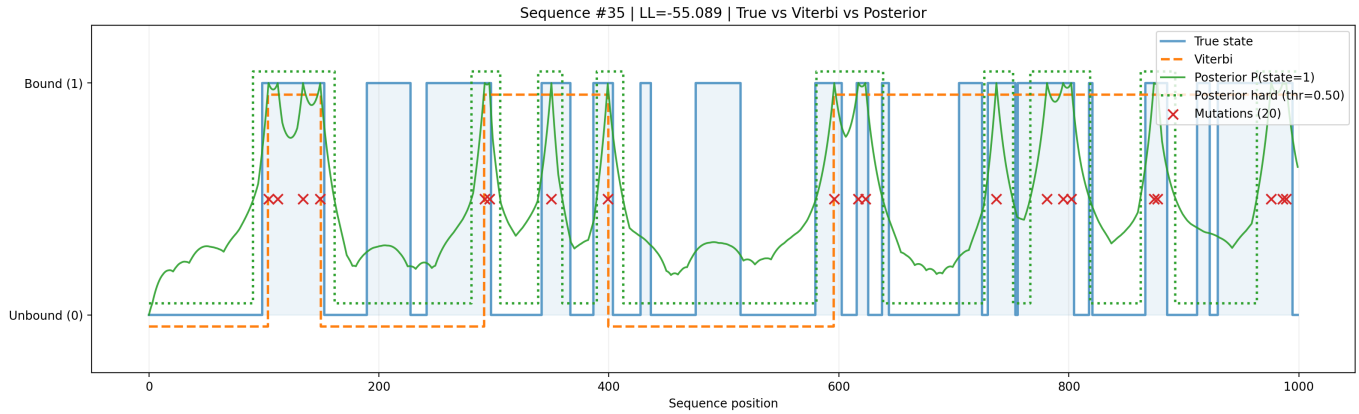

Fig. 1: Comparison of the ground truth hidden state (blue), the Viterbi decoded path (orange dashed), and the posterior probability of binding (green) for Sequence #35. Red crosses indicate observable mutations. The model successfully identifies binding events around mutation clusters.

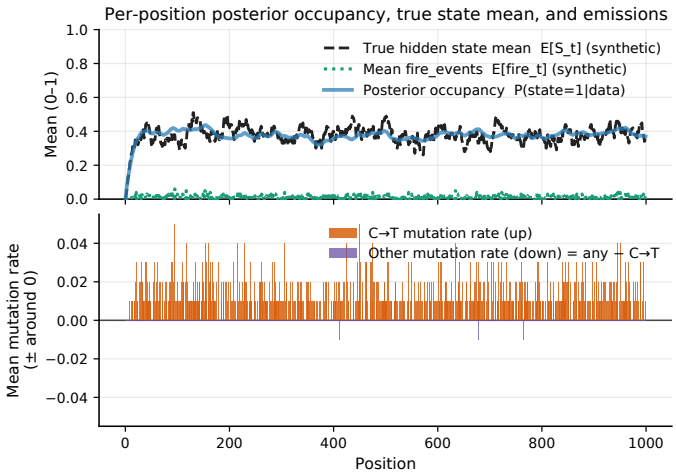

Fig. 2: Per-position statistics averaged across the synthetic dataset. Top: The posterior occupancy (blue) tracks the ground truth mean (black dashed). Bottom: The mutation rate profile showing C→T transitions (orange) vs. noise (purple).

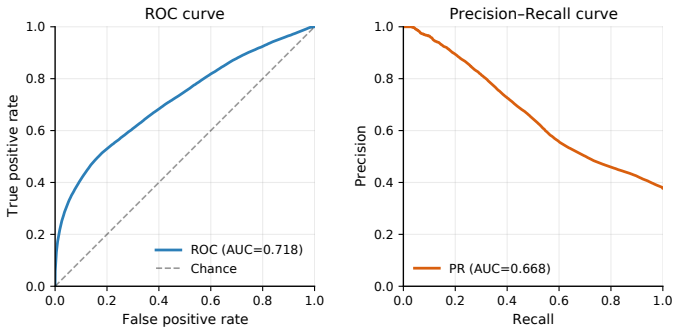

Fig. 3: Classification performance for hidden state recovery. Left: ROC curve (AUC = 0.718). Right: Precision-Recall curve (AUC = 0.668).

nature of the enzyme kinetic better. It represents the biological reality that the enzyme only binds for shorter stretches (accounts for the limited processivity).

VI. Conclusion

In this work, we proposed a hidden Markov framework to model the spatial activity of the Activation-Induced Cytidine Deaminase (AID) during Somatic Hypermutation. We assumed the enzyme's activity as a stochastic process with the latent states "bound" and "unbound". We implemented a generative model to simulate mutations on germline sequences, and applied Maximum Likelihood Estimation (MLE) to recover the kinetic parameters. Finally, we tried to recover the hidden binding paths using Viterbi decoding and posterior state marginalization.

We demonstrated the basic feasibility of this pipeline. The model was able to infer processivity and recruitment rates with reasonable accuracy and successfully identified binding events in regions of high mutational density. However, the current version shows strong limitations, particularly regarding the ability to identify the correct parameters in the face of sparse data.

Future work is needed to address these issues and to make the model usable for in vivo data. Specifically, we propose:

- **Parameter Decoupling:** To resolve the issue with finding the right parameters, future versions should fix well-known biological constants (such as mutation efficiency) based on experimental literature, which would allow the optimizer to focus solely on the unknown sequence biases that would then no longer be stuck in products with other parameters.
- **Model Complexity:** The current model simplifies the downstream repair mechanisms only focusing on C → T. A more elaborate model should explicitly account for the different error-prone repair pathways (BER vs. MMR), maybe using them as distinct emission states to achieve

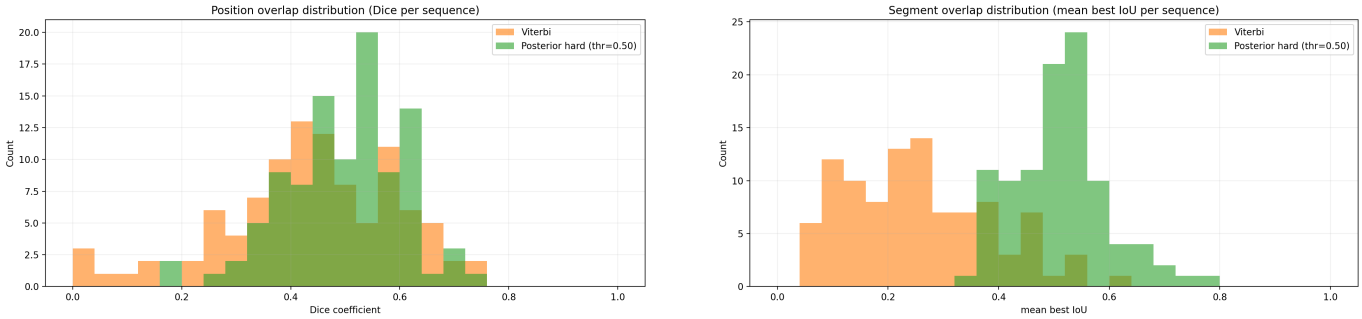

Fig. 4: Distribution of segmentation metrics per sequence. Left: Dice coefficient histogram. Right: Mean best Intersection over Union (IoU) per sequence. The hard-thresholded posterior (green) generally outperforms the Viterbi path (orange) in overlap consistency.

better resolution.

- **Data Scale and Real-World Application:** Increasing the dataset size could help stabilize the MLE by providing more mutation events. Further validation should move to semi-synthetic data (real germline sequences with synthetic mutations) to test the model with realistic sequence contexts (non-uniform), before eventually using the model for actual B-cell Receptor (BCR) sequencing data to infer real-world enzymatic dynamics.

References

[1] A. M. e. a. Shen HM, Poirier MG, "The activation-induced cytidine deaminase (aid) efficiently targets dna in nucleosomes but only during transcription." *J Exp Med*, 2009.

[2] G. Maul, "Aid and somatic hypermutation," *Adv Immunology*, 2010.

[3] G. M. Pham P, Afif S, "Structural analysis of the activation-induced deoxycytidine deaminase required in immunoglobulin diversification," *DNA Repair (Amst)*, 2016.

[4] G. M. Chelico L, Pham P, "Stochastic properties of processive cytidine dna deaminases aid and apobec3g," *Philos Trnas R Soc Lond B Biol*, 2009.

[5] B. Pilzecker and H. Jacobs, "Mutating for good: Dna damage responses during somatic hypermutation," *Frontiers in Immunology*, vol. Volume 10 - 2019, 2019. [Online]. Available: https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2019.00438

[6] B. J. J. M. e. a. Senavirathne, G., "Activation-induced deoxycytidine deaminase (aid) co-transcriptional scanning at single-molecule resolution," *Nat Commun 6*, 2015.

[7] W. Y. e. a. Taylor BJ, Nik-Zainal S, "Dna deaminases induce break-associated mutation showers with implication of apobec3b and 3a in breast cancer kataegis." *Elife*, 2013.

[8] A. S. G. M. Mak CH, Pham P, "A mathematical model for scanning and catalysis on single-stranded dna, illustrated with activation-induced deoxycytidine deaminase," *JBC Press*, 2013.

[9] M. C. G. M. Pham P, Malik S, "Aid–rna polymerase ii transcription-dependent deamination of igv dna," *Nucleic Acid Research*, 2019.

[10] V. R. e. a. Bartl B, Schoeberl UE, "Somatic hypermutation patterns are shaped by both motif position and sequence grammar," *EMBO J*, 2025.

[11] G. Yaari, J. Vander Heiden, M. Uduman, D. Gadala-Maria, N. Gupta, J. N. Stern, K. O'Connor, D. Hafler, U. Laserson, F. Vigneault, and S. Kleinstein, "Models of somatic hypermutation targeting and substitution based on synonymous mutations from high-throughput immunoglobulin sequencing data," *Frontiers in Immunology*, vol. Volume 4 - 2013, 2013. [Online]. Available: https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2013.00358

[12] T. M. Catherine Tang, Artem Krantsevich, "Deep learning model of somatic hypermutation reveals importance of sequence context beyond targeting of aid and pol hotspots," *bioRxiv*, 2021.

[13] D. Jurafsky and J. H. Martin, *Hidden Markov Models*. Unpublished draft manuscript, 2025, draft of August 24, 2025. [Online]. Available: https://web.stanford.edu/~jurafsky/slp3/

[14] SciPy Community, *SciPy Optimization — `scipy.optimize.minimize` with L-BFGS-B method*, SciPy, 2023, accessed: 2025-12-21. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/optimize.minimize-lbfgsb.html

[15] R. H. Byrd, P. Lu, and J. Nocedal, "A limited memory algorithm for bound constrained optimization," *SIAM Journal on Scientific and Statistical Computing*, vol. 16, no. 5, pp. 1190–1208, 1995.

[16] H. O. Aggrawal and J. Modersitzki, "Hessian initialization strategies for l-bfgs solving non-linear inverse problems," 2021. [Online]. Available: https://arxiv.org/abs/2103.10010